PART III : MATHEMATICS

SECTION 1 (Maximum Marks: 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : −1 In all other cases.

Q.37 Let $P=\begin{bmatrix}1 & 0 & 0\\ 4 & 1 & 0\\ 16 & 4 & 1\end{bmatrix}$ and I be the identity matrix of order 3. If $Q=[q_{ij}]$ is a matrix such that $P^{50}-Q=I$, then $\frac{q_{31}+q_{32}}{q_{21}}$ equals

(A) 52 (B) 103 (C) 201 (D) 205

Q.38 Let $b_i>1$ for $i=1,2,\ldots,101$. Suppose $\log_e b_1, \log_e b_2, \ldots, \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \ldots, a_{101}$ are in A.P. such that $a_1=b_1$ and $a_{51}=b_{51}$. If $t=b_1+b_2+\cdots+b_{51}$ and $s=a_1+a_2+\cdots+a_{51}$, then

(A) $s>t$ and $a_{101}>b_{101}$ (B) $s>t$ and $a_{101}<b_{101}$

(C) $s<t$ and $a_{101}>b_{101}$ (D) $s<t$ and $a_{101}<b_{101}$

Q.39 The value of $\sum_{k=1}^{13}\frac{1}{\sin\left(\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$ is equal to

(A) $3+\sqrt{3}$ (B) $2(3-\sqrt{3})$ (C) $2(\sqrt{3}-1)$ (D) $2(2+\sqrt{3})$

Space for rough work

Answers for the above questions

Ans for Q.37: (B)	Ans for Q.38: (B)	Ans for Q.39: (C)

Q.40 The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} dx$ is equal to

(A) $\frac{\pi^2}{4} - 2$ (B) $\frac{\pi^2}{4} + 2$ (C) $\pi^2 - e^{\frac{\pi}{2}}$ (D) $\pi^2 + e^{\frac{\pi}{2}}$

Q.41 Area of the region $\left\{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ is equal to

(A) $\frac{1}{6}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

Q.42 Let P be the image of the point $(3, 1, 7)$ with respect to the plane $x - y + z = 3$. Then the equation of the plane passing through P and containing the straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}$ is

(A) $x + y - 3z = 0$ (B) $3x + z = 0$

(C) $x - 4y + 7z = 0$ (D) $2x - y = 0$

Space for rough work

Answers for the above questions

Ans for Q.40: (A)	Ans for Q.41: (C)	Ans for Q.42: (C)

SECTION 2 (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks	: +4	If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.
Partial Marks	: +1	For darkening a bubble corresponding **to each correct option**, provided NO incorrect option is darkened.
Zero Marks	: 0	If none of the bubbles is darkened.
Negative Marks	: –2	In all other cases.

- For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened.

Q.43 Let $f(x)=\lim\limits_{n\to\infty}\left(\dfrac{n^n(x+n)(x+\frac{n}{2})\cdots(x+\frac{n}{n})}{n!\,(x^2+n^2)(x^2+\frac{n^2}{4})\cdots(x^2+\frac{n^2}{n^2})}\right)^{\frac{x}{n}}$, for all $x>0$. Then

(A) $f\left(\frac{1}{2}\right)\geq f(1)$

(B) $f\left(\frac{1}{3}\right)\leq f\left(\frac{2}{3}\right)$

(C) $f'(2)\leq 0$

(D) $\dfrac{f'(3)}{f(3)}\geq\dfrac{f'(2)}{f(2)}$

Q.44 Let $a,b\in\mathbb{R}$ and $f:\mathbb{R}\to\mathbb{R}$ be defined by $f(x)=a\cos(|x^3-x|)+b|x|\sin(|x^3+x|)$. Then f is

(A) differentiable at $x=0$ if $a=0$ and $b=1$

(B) differentiable at $x=1$ if $a=1$ and $b=0$

(C) **NOT** differentiable at $x=0$ if $a=1$ and $b=0$

(D) **NOT** differentiable at $x=1$ if $a=1$ and $b=1$

Space for rough work

Answers for the above questions

Ans for Q.43: (B) and (C)	Ans for Q.44: (A) and (B)

Q.45 Let $f:\mathbb{R}\to(0,\infty)$ and $g:\mathbb{R}\to\mathbb{R}$ be twice differentiable functions such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $f'(2)=g(2)=0$, $f''(2)\neq 0$ and $g'(2)\neq 0$. If $\lim\limits_{x\to 2}\dfrac{f(x)g(x)}{f'(x)g'(x)}=1$, then

(A) f has a local minimum at $x=2$

(B) f has a local maximum at $x=2$

(C) $f''(2)>f(2)$

(D) $f(x)-f''(x)=0$ for at least one $x\in\mathbb{R}$

Q.46 Let $f:\left[-\dfrac{1}{2},2\right]\to\mathbb{R}$ and $g:\left[-\dfrac{1}{2},2\right]\to\mathbb{R}$ be functions defined by $f(x)=[x^2-3]$ and $g(x)=|x|f(x)+|4x-7|f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y\in\mathbb{R}$. Then

(A) f is discontinuous exactly at three points in $\left[-\dfrac{1}{2},2\right]$

(B) f is discontinuous exactly at four points in $\left[-\dfrac{1}{2},2\right]$

(C) g is **NOT** differentiable exactly at four points in $\left(-\dfrac{1}{2},2\right)$

(D) g is **NOT** differentiable exactly at five points in $\left(-\dfrac{1}{2},2\right)$

Space for rough work

Answers for the above questions

Ans for Q.45: (A) and (D)	Ans for Q.46: (B) and (C)

Q.47 Let $a, b \in \mathbb{R}$ and $a^2 + b^2 \neq 0$. Suppose $S = \left\{ z \in \mathbb{C} : z = \frac{1}{a + ibt}, t \in \mathbb{R}, t \neq 0 \right\}$, where $i = \sqrt{-1}$. If $z = x + iy$ and $z \in S$, then (x, y) lies on

(A) the circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0 \right)$ for $a > 0, b \neq 0$

(B) the circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a}, 0 \right)$ for $a < 0, b \neq 0$

(C) the x-axis for $a \neq 0, b = 0$

(D) the y-axis for $a = 0, b \neq 0$

Q.48 Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then

(A) $SP = 2\sqrt{5}$

(B) $SQ : QP = (\sqrt{5} + 1) : 2$

(C) the x-intercept of the normal to the parabola at P is 6

(D) the slope of the tangent to the circle at Q is $\frac{1}{2}$

Space for rough work

Answers for the above questions

Ans for Q.47: (A), (C) and (D)	Ans for Q.48: (A), (C) and (D)

Q.49 Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$
$$3x - 2y = \mu$$

Which of the following statement(s) is(are) correct?

(A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

Q.50 Let $\hat{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ be a unit vector in $\mathbb{R}^3$ and $\hat{w} = \frac{1}{\sqrt{6}}(\hat{i} + \hat{j} + 2\hat{k})$. Given that there exists a vector $\vec{v}$ in $\mathbb{R}^3$ such that $\left|\hat{u} \times \vec{v}\right| = 1$ and $\hat{w} \cdot (\hat{u} \times \vec{v}) = 1$. Which of the following statement(s) is(are) correct?

(A) There is exactly one choice for such $\vec{v}$

(B) There are infinitely many choices for such $\vec{v}$

(C) If $\hat{u}$ lies in the xy-plane then $|u_1| = |u_2|$

(D) If $\hat{u}$ lies in the xz-plane then $2|u_1| = |u_3|$

Space for rough work

Answers for the above questions

Ans for Q.49: (B), (C) and (D)	Ans for Q.50: (B) and (C)

SECTION 3 (Maximum Marks: 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

  Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

  Zero Marks : 0 In all other cases.

PARAGRAPH 1

Football teams T_1 and T_2 have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of T_1 winning, drawing and losing a game against T_2 are $\frac{1}{2}, \frac{1}{6}$ and $\frac{1}{3}$, respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams T_1 and T_2, respectively, after two games.

Q.51 $P(X > Y)$ is

(A) $\frac{1}{4}$ (B) $\frac{5}{12}$ (C) $\frac{1}{2}$ (D) $\frac{7}{12}$

Q.52 $P(X = Y)$ is

(A) $\frac{11}{36}$ (B) $\frac{1}{3}$ (C) $\frac{13}{36}$ (D) $\frac{1}{2}$

Space for rough work

Answers for the above questions

Ans for Q.51: (B)	Ans for Q.52: (C)

PARAGRAPH 2

Let $F_1(x_1, 0)$ and $F_2(x_2, 0)$, for $x_1 < 0$ and $x_2 > 0$, be the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{8} = 1$. Suppose a parabola having vertex at the origin and focus at F_2 intersects the ellipse at point M in the first quadrant and at point N in the fourth quadrant.

Q.53 The orthocentre of the triangle F_1MN is

(A) $\left(-\frac{9}{10}, 0\right)$ (B) $\left(\frac{2}{3}, 0\right)$ (C) $\left(\frac{9}{10}, 0\right)$ (D) $\left(\frac{2}{3}, \sqrt{6}\right)$

Q.54 If the tangents to the ellipse at M and N meet at R and the normal to the parabola at M meets the x-axis at Q, then the ratio of area of the triangle MQR to area of the quadrilateral MF_1NF_2 is

(A) 3 : 4 (B) 4 : 5 (C) 5 : 8 (D) 2 : 3

END OF THE QUESTION PAPER

Space for rough work

Answers for the above questions

Ans for Q.53: (A)	Ans for Q.54: (C)